neutrons released from one fission will hit other uranium or plutonium nuclei and fission them, producing more neutrons that fission more nuclei, producing still more neutrons that fission still more nuclei, and so on. The result of this chain reaction, if uncontrolled, is a huge explosion (an atomic bomb blast); if controlled in a reactor, the result can be highly efficient electric power.

Bohr and Wheeler were remarkably successful. They discovered how the laws of physics produce fission, and they predicted which nuclei would be the most effective at sustaining chain reactions: uranium-235 (which would become the fuel for the bomb to destroy Hiroshima) and plutonium-239 (a type of nucleus that does not exist in nature but that the American physicists would soon learn how to make in nuclear reactors and would use to fuel the bomb to destroy Nagasaki). However, Bohr and Wheeler were not thinking of bombs in 1939; they only wanted to understand.

The Bohr–Wheeler article explaining nuclear fission was published in the same issue of the *Physical Review* as the Oppenheimer–Snyder article describing the implosion of a star. The publication date was 1 September 1939, the very day that Hitler's troops invaded Poland, triggering World War 11.

Yakov Borisovich Zel'dovich was bom into a Jewish family in Minsk in 1914; later that year his family moved to Saint Petersburg (renamed Leningrad in the 1920s, then restored to Saint Petersburg in the 1990s). Zel'-dovich completed high school at age fifteen and then, instead of entering university, went to work as a laboratory assistant at the Physicotechnical Institute in Leningrad. There he taught himself so much physics and chemistry and did such impressive research that, without any formal university training, he was awarded a Ph.D. in 1934, at age twenty.

In 1939, while Wheeler and Bohr were developing the theory of nuclear fission, Zel'dovich and a close friend, Yuli Borisovich Khariton, were developing the theory of chain reactions produced by nuclear fission: Their research was triggered by an intriguing (incorrect) suggestion from French physicist Francis Perrin that volcanic eruptions might be powered by natural, underground nuclear explosions, which result from a chain reaction of fissions of atomic nuclei. However, nobody including Perrin had worked